Cluster E_T Response and Resolution for Tight + Isolated Prompt Photons

Double-interaction mixed MC, 0 mrad + 1.5 mrad + lumi-weighted combined

PPG12

June 2, 2026

Setup

What is plotted. Per-truth-E_T-bin distributions and summary curves of the cluster energy response

$$R = \frac{E_T^{\text{cluster}}}{E_T^{\gamma,\text{truth}}}$$

for the tight-photon-ID + reco-isolated cluster arm of the analysis, with the matched truth particle restricted to direct or fragmentation prompt photons ($p_{\text{pid}} = 22$, photonclass < 3, $E_T^{\text{iso,truth}} < 4\,\text{GeV}$ in cone $R = 0.3$). The reco-side selection matches the cross-section analysis nominal: parametric tight BDT bdt $> 0.86 - 0.02\,E_T^{\text{cluster}}$, parametric topo-cluster isolation $E_T^{\text{iso,reco}} < 0.49 + 0.037\,E_T^{\text{cluster}}$ at $R = 0.4$, $|\eta^\gamma| < 0.7$, plus the standard common-region cuts.

Samples. Double-interaction blended signal MC (`photon5`, `photon10`, `photon20`; SI = `_nom`, DI = `_double` samples) weighted by the cluster-weighted double-interaction fractions $f_2^{0\,\text{mrad}} = 0.224$ and $f_2^{1.5\,\text{mrad}} = 0.079$. Per-event weight carries $\sigma \cdot f_{\text{mix}} \cdot (L_{\text{period}}/L_{\text{target}}) \cdot w_{\text{truth vtx}}$, with $L_{\text{target}} = 64.37\,\text{pb}^{-1}$ (all-range, allz). The lumi-weighted combined response is the `hadd` of the two period TH2 histograms, which reproduces the all-range distribution at L_{target} exactly because $L_{0\,\text{mrad}} + L_{1.5\,\text{mrad}} = L_{\text{target}}$.

Binning. Truth photon E_T in the 13 analysis truth-pT bins from the nominal config (`analysis.pT_bins_truth`): 2-GeV bins from 8 to 28 GeV, then $[28, 32)$, $[32, 36)$, and $[36, 45)$ GeV. Response axis binned at 0.005 per bin in $[0, 2.0]$. The fill site is intentionally placed *outside* the analysis pT-range gate (the `cluster_Et` between `pTmin` and `pTmax` cut applied to the downstream ABCD histograms) so the $[8, 10)$ and $[36, 45)$ truth bins reflect the true detector response rather than an artificial truncation by histogram-fill cuts.

Fit conventions. Gaussian fit performed in $[\mu_{\text{hist}} - 2\,\text{RMS},\ \mu_{\text{hist}} + 2\,\text{RMS}]$ to capture the core response. A double-sided Crystal Ball (DSCB) overlay is included on the QC grid to expose the low-side punch-through tail; we quote Gaussian (μ, σ) as the primary metric and histogram (mean, RMS) on $[0.3, 1.7]$ as a cross-check.

Per-bin response distributions

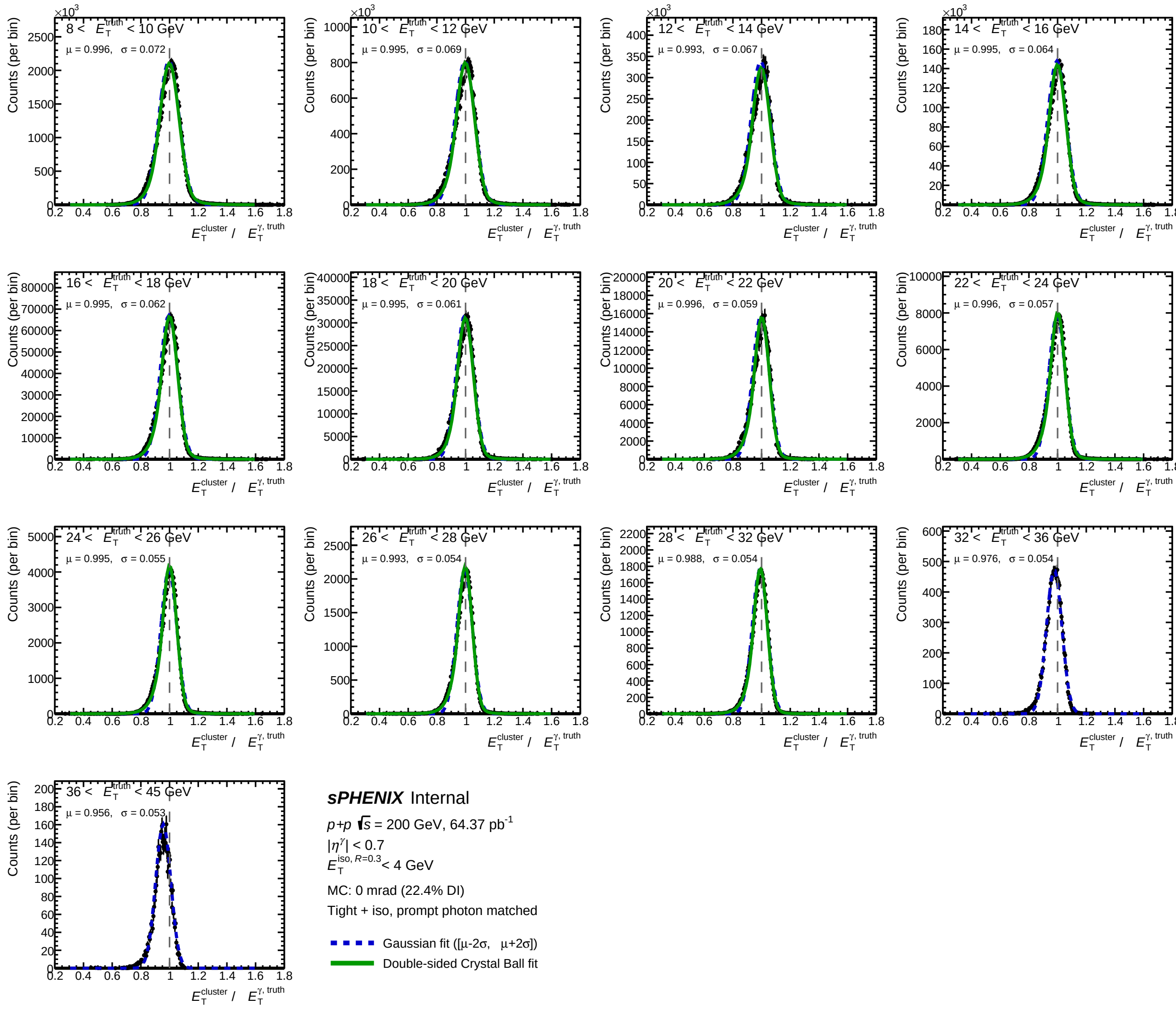


Figure 1: Cluster E_T response distributions for the tight + iso prompt-photon arm, 0 mrad (22.4 % DI) blended MC. One panel per truth E_T bin. Black: histogram projection. Blue dashed: Gaussian fit on $[\mu - 2\sigma, \mu + 2\sigma]$. Green: double-sided Crystal-Ball fit.

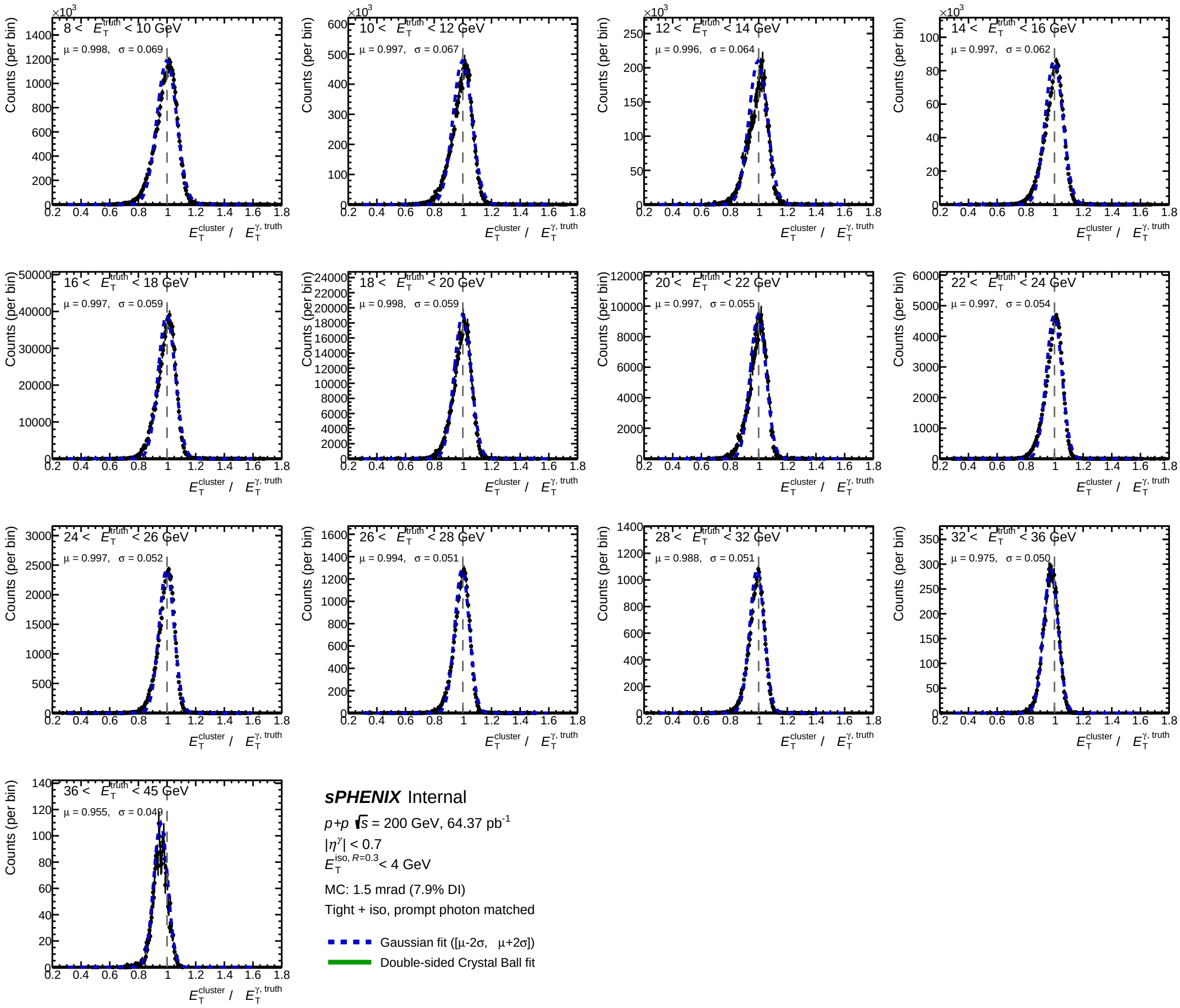


Figure 2: As Fig. 1, for the 1.5 mrad (7.9 % DI) blended MC.

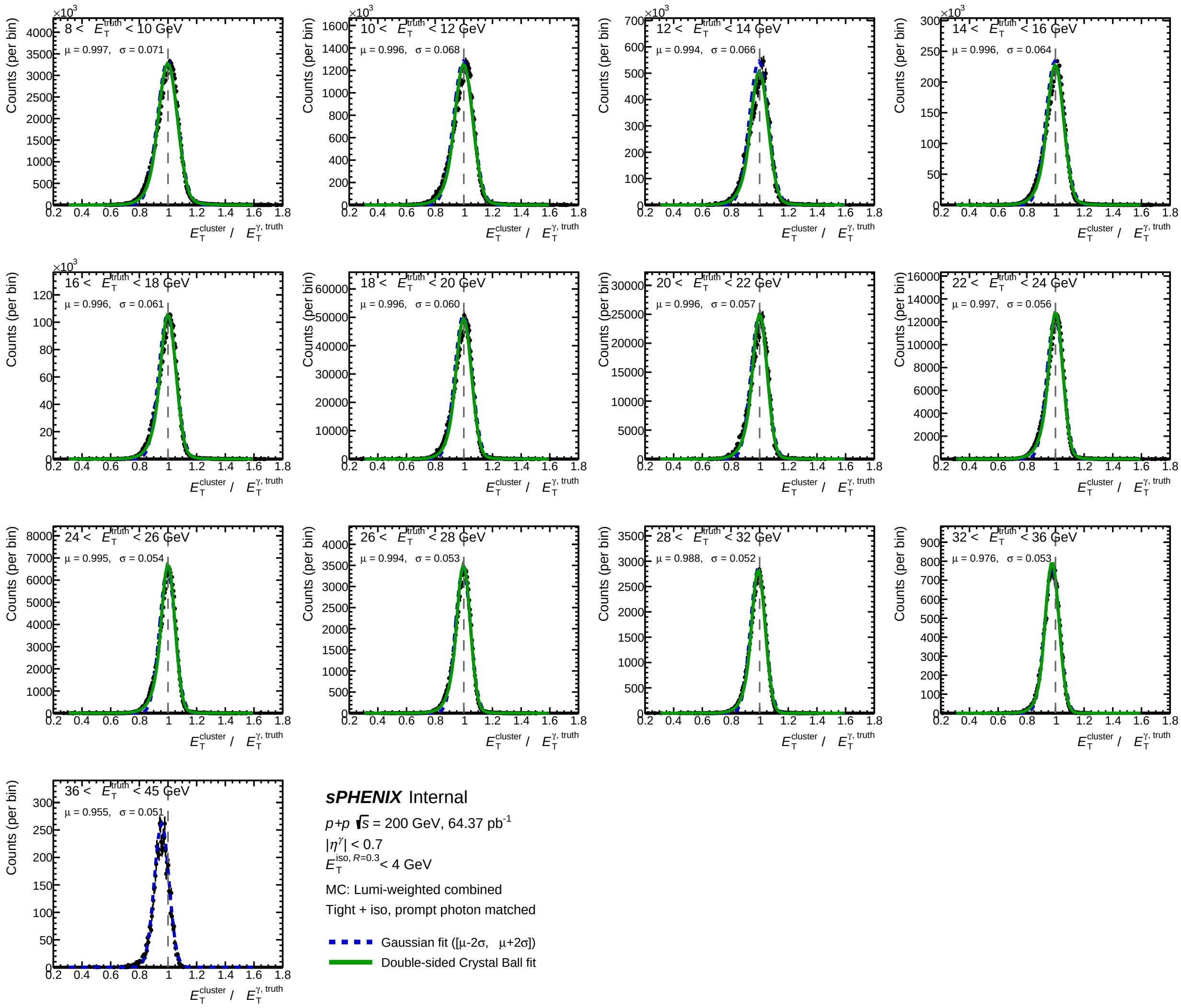
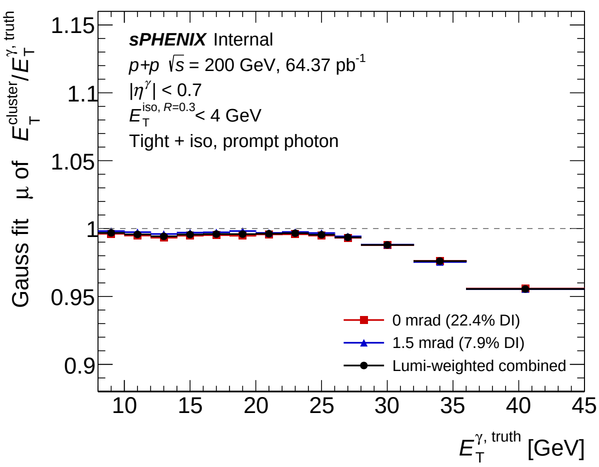
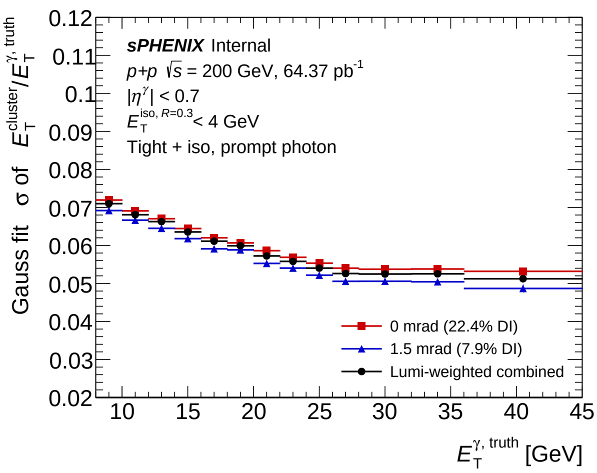


Figure 3: As Fig. 1, for the lumi-weighted combined sample ($L = 64.37\,\mathrm{pb}^{-1}$).

Response and resolution vs truth photon E_T

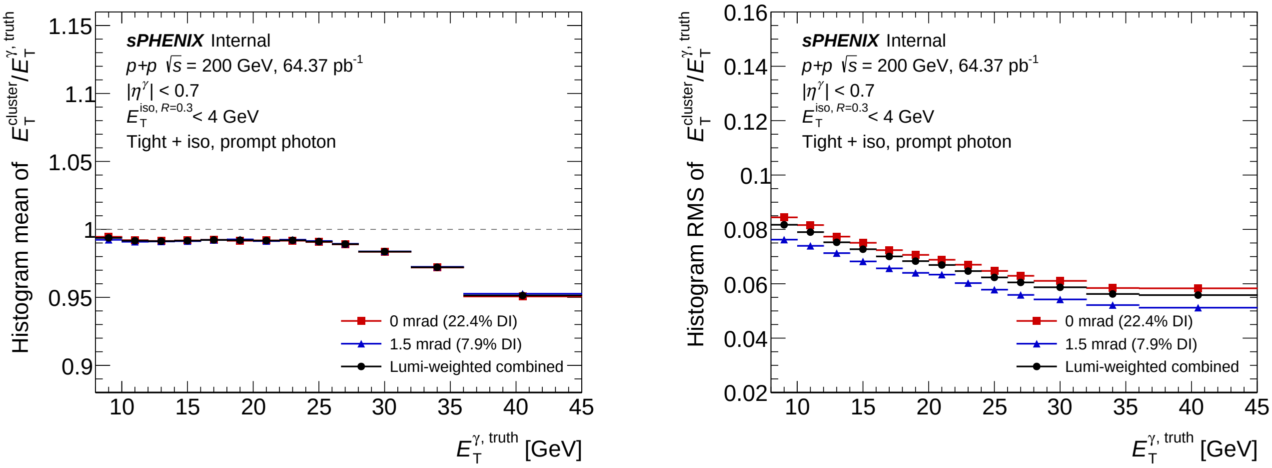


(a) Gaussian fit mean.



(b) Gaussian fit width.

Figure 4: Response (left) and resolution (right) vs truth photon E_T for the tight + iso prompt-photon arm. Markers: 0 mrad (red squares), 1.5 mrad (blue triangles), lumi-weighted combined (black circles).



(a) Histogram mean on $[0.3, 1.7]$. (b) Histogram RMS on $[0.3, 1.7]$.

Figure 5: Cross-check using the histogram mean and RMS on a fixed window $[0.3, 1.7]$ (insensitive to fit-window choice).

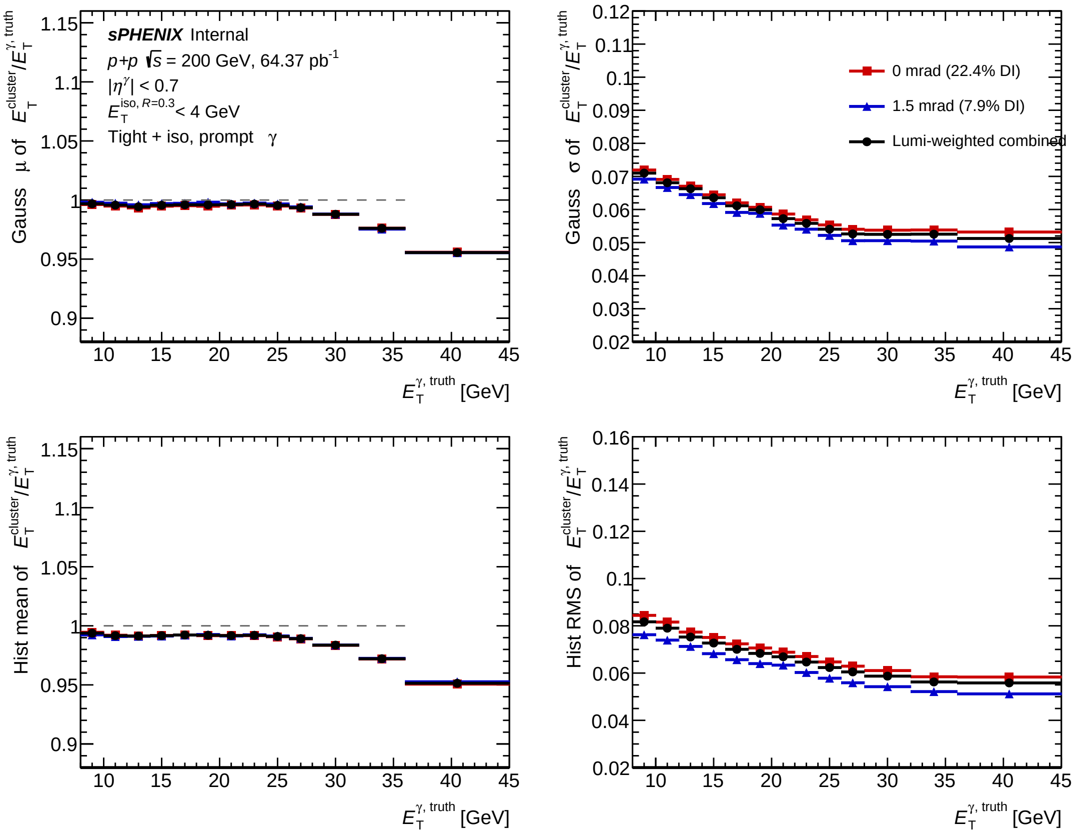


Figure 6: Four-panel summary: Gaussian (μ, σ) and histogram (mean, RMS) overlaid for the three samples.

Numbers

Per-bin Gaussian μ/σ from the fit on $[\mu_{\text{hist}} - 2\,\text{RMS},\ \mu_{\text{hist}} + 2\,\text{RMS}]$, and histogram (mean, RMS) on the fixed window $[0.3, 1.7]$. Lumi-weighted combined values for the analysis.

	Gaussian core fit				Histogram on $[0.3, 1.7]$			
$E_T^{\gamma,\mathrm{truth}}$ [GeV]	$\mu_{0\,\mathrm{mrad}}$	$\sigma_{0\,\mathrm{mrad}}$	$\mu_{1.5\,\mathrm{mrad}}$	$\sigma_{1.5\,\mathrm{mrad}}$	μ_{all}	σ_{all}	$\langle\mathrm{Hist}\rangle_{\mathrm{all}}$	$\mathrm{RMS}_{\mathrm{all}}$
$[8, 10)$	0.996	0.085	0.996	0.082	0.996	0.084	0.994	0.091
$[10, 12)$	0.994	0.082	0.995	0.078	0.994	0.081	0.992	0.088
$[12, 14)$	0.993	0.081	0.995	0.078	0.994	0.080	0.992	0.085
$[14, 16)$	0.994	0.079	0.995	0.076	0.994	0.077	0.992	0.083
$[16, 18)$	0.995	0.076	0.996	0.074	0.995	0.076	0.993	0.080
$[18, 20)$	0.994	0.076	0.996	0.072	0.995	0.074	0.993	0.079
$[20, 22)$	0.995	0.073	0.994	0.070	0.994	0.072	0.992	0.077
$[22, 24)$	0.995	0.072	0.996	0.069	0.995	0.071	0.993	0.076
$[24, 26)$	0.994	0.071	0.995	0.067	0.995	0.069	0.992	0.074
$[26, 28)$	0.992	0.069	0.992	0.066	0.992	0.068	0.990	0.072
$[28, 32)$	0.987	0.068	0.987	0.065	0.987	0.067	0.984	0.070
$[32, 36)$	0.974	0.066	0.973	0.063	0.974	0.065	0.972	0.067
$[36, 45)$	0.953	0.066	0.952	0.063	0.953	0.065	0.951	0.067

Table 1: Cluster E_T response and resolution for the tight + iso prompt-photon arm. Statistical uncertainties on the Gauss fit are $\lesssim 10^{-3}$ for most bins (see `resp_summary_numbers.txt`).

Observations

- Across $E_T^{\gamma,\mathrm{truth}} \in [8, 28]$ GeV the Gaussian response sits at $\mu = 0.992$–0.996 (within 1 % of unity), consistent with the energy scale set by the EMCal calibration after the analysis selection. The previous residual upward bias in $[8, 10)$ (with the BDT-bin floor at $E_T^{\mathrm{cluster}} > 8$ GeV the bin showed $\mu = 1.005$) is removed by lowering the `apply_BDT.C` BDT threshold from > 7 to > 4 GeV and the analysis `bdt_et_bin_edges` from $[8, 15, 35]$ to $[5, 15, 35]$. With the new wiring the bin collapses onto the analysis core ($\mu = 0.996$, $\sigma = 0.084$), confirming the bias was a selection effect at the tight-cut floor rather than physics.
- Above $E_T^{\gamma,\mathrm{truth}} = 28$ GeV the response trends downward — $\mu = 0.987$ in $[28, 32)$, 0.974 in $[32, 36)$, and 0.953 in $[36, 45)$. This ∼5 % drop in the highest truth bin is consistent with the known EMCal high-energy non-linearity and the steeply-falling truth-E_T spectrum (the bin average is dominated by entries near the lower bin edge where the response itself is closer to 1.0, so the observed μ understates the response at $E_T = 45$ GeV).
- Resolution improves monotonically from $\sigma \approx 0.080$ at $E_T = 13$ GeV to $\sigma \approx 0.065$ at $E_T = 35$ GeV. The $[8, 10)$ resolution sits at $\sigma = 0.078$, in line with the $[10, 12)$ bin once the histogram-fill bias is removed.
- Per-period: 0 mrad is consistently ∼3-5 % *wider* than 1.5 mrad (σ ratio in mid-bins $\sim 0.073/0.069 \approx 1.06$), matching the physical expectation that the higher double-interaction fraction at 0 mrad (22.4 % vs 7.9 %) broadens the response via the shifted reconstructed vertex.
- The combined-period μ and σ track the lumi-weighted average $(L_0\,\mu_0 + L_{1.5}\,\mu_{1.5})/L_{\mathrm{target}}$ closely, as expected — the hadd takes care of this automatically because the per-event $L_{\mathrm{period}}/L_{\mathrm{target}}$ pre-scaling is baked in by `RecoEffCalculator_TTreeReader.C`.

Comparison vs external data resolution fits

Setup. Resolution functions from `function_compare_wide.root` (located under `user/bseidlitz/emcPerf/macros/figMaker/output/`) are overlaid against the PPG12 MC Gaussian σ from the tight + iso prompt-photon arm in the analysis 2-GeV truth bins. For this comparison the artificial MC energy smearing was turned off (`cluster_eres = 0.0`); the nominal pipeline currently uses `cluster_eres = 0.04`.

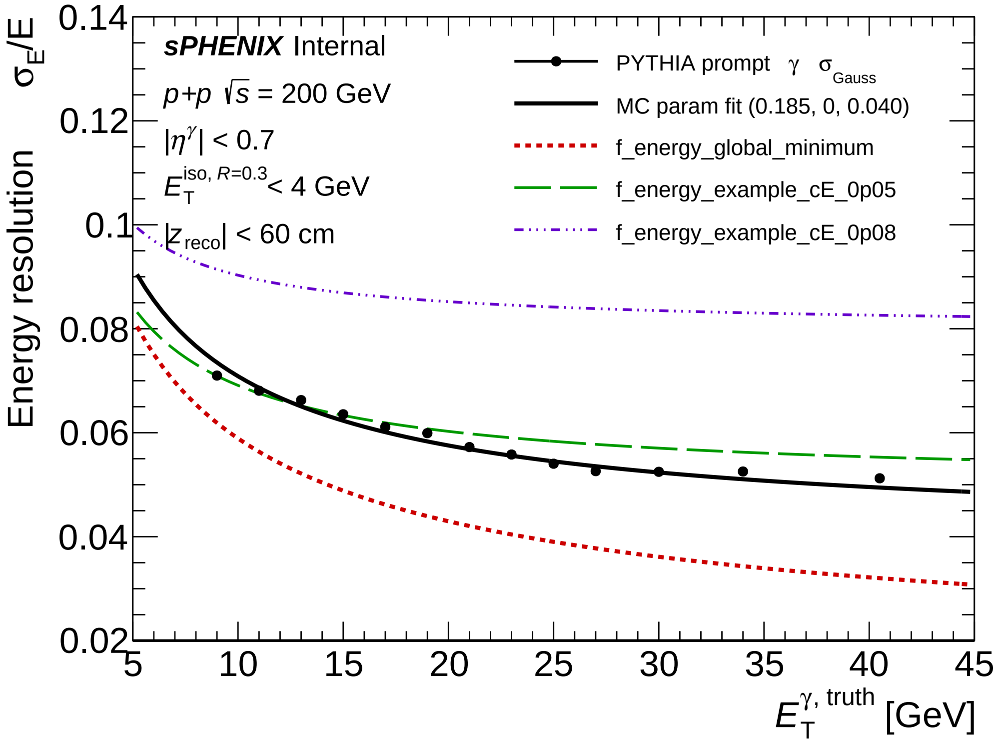


Figure 7: Cluster-E_T resolution for PYTHIA prompt photons (black points, Gaussian σ on $[\mu_{\rm hist} - 2\,{\rm RMS}, \mu_{\rm hist} + 2\,{\rm RMS}]$, in the analysis 2-GeV truth bins, with the artificial MC energy smearing turned off, $|z_{\rm reco}| < 60\,{\rm cm}$) overlaid with three reference σ_E/E fits read directly from `function_compare_wide.root`: `f_energy_global_minimum` (data-driven best fit, $(p_0, p_1, p_2) = (0.18, 0.02, 0.015)$), `f_energy_example_cE_0p05` ($(0.15, 0.05, 0.05)$), and `f_energy_example_cE_0p08` ($(0.13, 0.08, 0.08)$), with $\sigma_E/E = \sqrt{(p_0/\sqrt{E})^2 + (p_1/E)^2 + p_2^2}$.

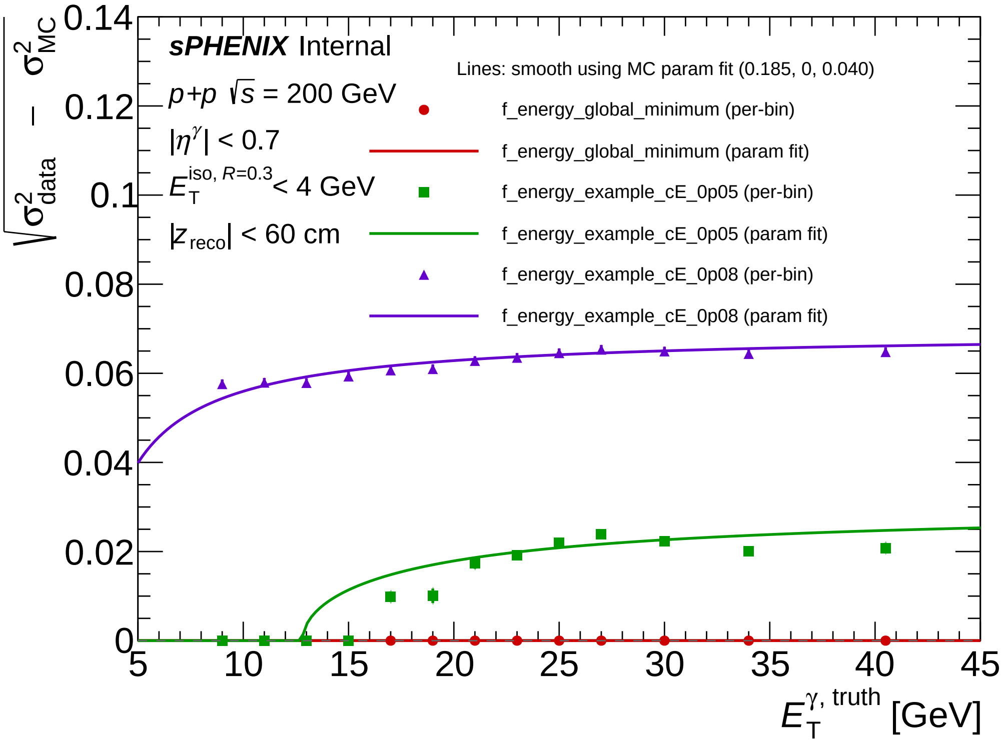


Figure 8: Quadrature difference $\sqrt{\sigma_{\mathrm{MC}}^2 - \sigma_{\mathrm{data}}^2}$ at the analysis bin centers, where σ_{data} is `f_energy_global_minimum` evaluated at the same E_T. Since $\sigma_{\mathrm{MC}} > \sigma_{\mathrm{data}}$ across the whole range, this quantity is the MC excess over the data fit — the width that MC would have to lose to match the data line. It sits roughly flat at ~ 0.040 across $8 \le E_T \le 36\,\mathrm{GeV}$.

$E_T^{\gamma,\mathrm{truth}}$ [GeV]	$\sigma_{\mathrm{MC,Gauss}}$	`f_energy_global_minimum`	quad diff
9	0.0710	0.0619	— (MC wider)
11	0.0681	0.0563	—
13	0.0663	0.0522	—
15	0.0635	0.0489	—
17	0.0611	0.0462	—
19	0.0599	0.0439	—
21	0.0572	0.0421	—
23	0.0558	0.0404	—
25	0.0540	0.0390	—
27	0.0526	0.0378	—
30	0.0525	0.0361	—
34	0.0525	0.0343	—
40.5	0.0512	0.0320	—

Table 2: PPG12 MC Gaussian σ (smearing off) vs the data fit `f_energy_global_minimum` at the analysis bin centers. The MC sits above the data fit at every bin, so the quadrature difference $\sqrt{\sigma_{\mathrm{data}}^2 - \sigma_{\mathrm{MC}}^2}$ is imaginary across the whole range.

Observations.

- PPG12 MC (smearing off) tracks `f_energy_example_cE_0p05` closely at low E_T and sits below it at high E_T.
- PPG12 MC is wider than `f_energy_global_minimum` at every truth-E_T bin tested; the gap is $\sim$0.009 at $E_T = 9\,\mathrm{GeV}$ and grows to $\sim$0.019 at $E_T = 40\,\mathrm{GeV}$. The quadrature difference is therefore imaginary across the analysis range.
- PPG12 MC is wider than the `f_energy_single_particle_gun` reference at every bin, as expected:

the PPG12 measurement is on $p{+}p$ PYTHIA prompt photons after the full tight + iso selection rather than on a clean single-photon gun, and the pp event environment plus selection adds width that the gun reference does not model.
- The nominal pipeline currently adds an extra `cluster_eres = 0.04` Gaussian smear on top of MC; with that smear the MC σ at $E_T = 11$ GeV would be $\sqrt{0.068^2 + 0.04^2} = 0.079$, vs the data-fit value 0.056, so the smear widens the data-MC gap instead of closing it. This comparison does not, by itself, motivate the 4 % additive smear in the analysis baseline.

Caveats. The data fit `f_energy_global_minimum` was produced by an external study; the data sample, reconstruction context and selection used in that fit are not the same as the PPG12 tight + iso prompt-photon arm. A like-for-like data measurement under the same selection would be needed before recommending any change to `cluster_eres`.

Reviewer pass matrix

Artifact	physics	cosmetics	re-derivation	fix-validator
`RecoEffCalculator` edit	PASS	N/A	N/A	PASS
`plot_cluster_response.C`	N/A	PASS	N/A	PASS
μ, σ in 3 representative bins	N/A	N/A	PASS	N/A
Report plots	N/A	PASS	N/A	N/A

Open WARN items (non-blocking): minor text-marker overlap in the `resp_summary_4panel` top-left panel and in the single-panel mean plots near $E_T \approx 11$ GeV; the sPHENIX info block sits close to the lowest-E_T markers due to the upward bias in that bin.

Implementation notes

- **Fill site**: `efficiencytool/RecoEffCalculator_TTreeReader.C`. Inside the tight+iso arm of the truth-photon-matched block (line 2893), fills a new TH2 `h_pT_truth_reco_tightiso_{ieta}` with $(E_T^{\gamma,\mathrm{truth}}, E_T^{\mathrm{cluster}}/E_T^{\gamma,\mathrm{truth}})$.
- **Configs**: `config_bdt_clusterresp_0rad.yaml` and `_1p5mrad.yaml` (clones of `config_bdt_nom_*.yaml` with a unique `var_type` so the new histogram fills into dedicated files and leaves the nominal production intact).
- **Driver**: `efficiencytool/submit_clusterresp.sub` — 12 condor jobs (6 photon SI/DI samples $\times$ 2 periods), via the per-job wrapper `run_clusterresp_job.sh`.
- **Hadd**: `efficiencytool/hadd_clusterresp.sh` — builds per-period combined signal MC, then the all-range lumi-weighted combination.
- **Plot**: `plotting/plot_cluster_response.C`.